L3: Precomputation and multi-target attacks on fixed curves

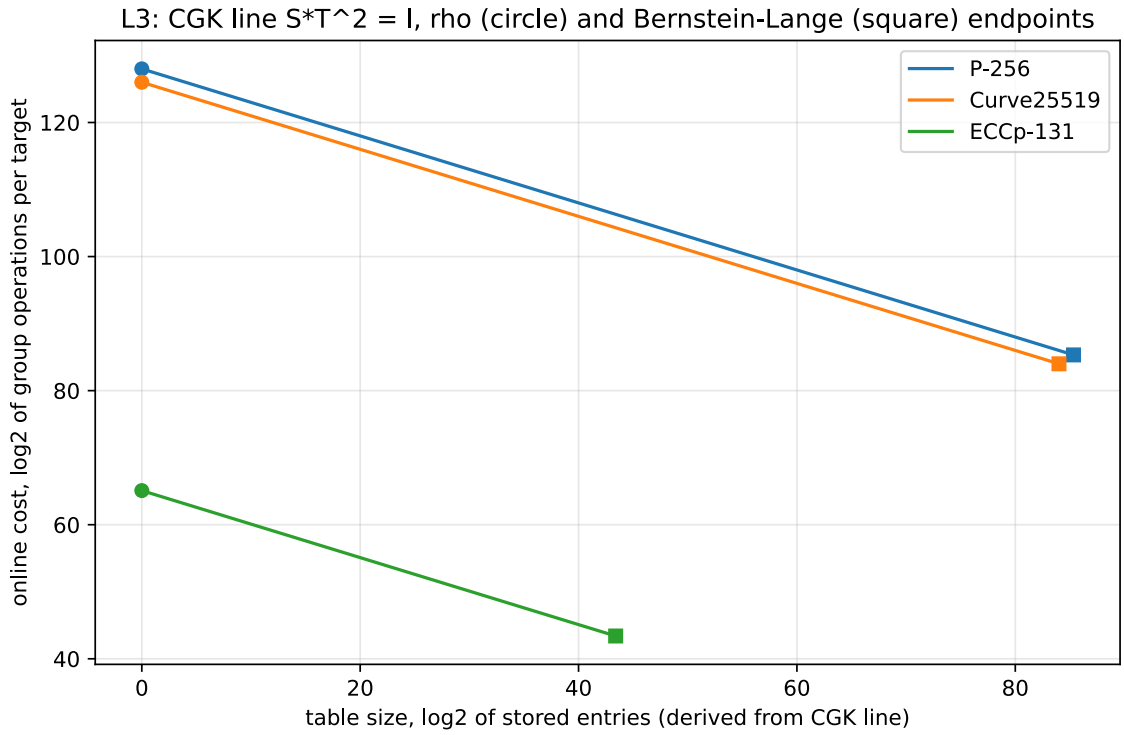
Writeup, derived from published bounds and atlas cards `KR-RHO-ea34b8` and `KR-RHO-46c2c6`. No measurement. Units are log2 of group operations (or of stored entries). Not a break.

Implied exponents

Curve	rho / target	BL table	BL precomp	BL online / target	BL online, k=2^10	KS total, k=2^10	KS / target
P-256 (log2 l = 256.00)	128.00	85.33	170.67	85.33	95.33	133.50	123.50
Curve25519 (252.00)	126.00	84.00	168.00	84.00	94.00	131.50	121.50
ECCp-131 (130.19)	65.09	43.40	86.79	43.40	53.40	70.60	60.60

BL = Bernstein–Lange; KS = Kuhn–Struik total √(2kl). The task's "table of about l^(2/3)" is split here: stored table about l^(1/3) entries, precomputation about l^(2/3) operations.

Tradeoff



Derived from the Corrigan-Gibbs–Kogan line S·T² = l. Rho and Bernstein–Lange are the endpoints; intermediate points are not attained algorithms.

Reading

The sub-rho online cost is paid by a precomputation far above rho (about 2^42.7 for P-256). It is not an attack on any of the three curves.